Differential Splicing with Subisoform Covariance

Overview

Questions and unit of analysis

The analysis asks two distinct questions:

1. Does local splice-path usage differ by experimental condition within a cell type?

2. Does local splice-path usage differ across cell types within the same mice?

The primary analysis aggregates cells into *mouse-by-condition-by-cell-type pseudobulks*. An evaluated alternative retains each cell's EC counts and transcript baseline in the local likelihood but estimates one shared path effect for the pseudobulk. Both approaches keep biological replication at the mouse level; cells are never treated as independent condition replicates.

A splice block is a local region between constitutive exonic anchors. Transcripts that take the same path through that region are collapsed to one subisoform. This targets local splicing choices, which can be identifiable even when the corresponding full-length transcripts are not.

For condition effects, each hypothesis is indexed by a splice block and a cell type. It is an omnibus test of

$$H_0 : \quad \text{mean path usage is equal across the included conditions.}$$

Only conditions represented by at least three mouse pseudobulks are included. With two conditions and two paths, this is a precision-weighted differential-PSI test. With more conditions or paths, all condition coefficients and path log-ratios are tested jointly.

For cell-type effects, each hypothesis is indexed only by a nonredundant splice-path partition. It is an omnibus test of equal path usage across the included cell types, after absorbing mouse and condition into a mouse-specific fixed effect. Cell types must be represented in at least three mice, and the primary FDR family requires at least 30 pseudobulk observations per partition. These two hypothesis families are reported separately.

Workflow

The analysis proceeds in five steps:

1. Construct local splice blocks from the transcript annotation and collapse transcripts to paths within each block.

2. Aggregate paired oligo(dT) and random-hexamer EC counts, and hierarchical transcript estimates, within each mouse, condition, and cell type.

3. Refine path usage in each pseudobulk with the exact paired-primer EC likelihood. Optionally, retain separate cell likelihood terms while sharing one pseudobulk path perturbation.

4. Estimate path-logratio covariance from EC ambiguity and UMI sampling. Estimates with weak information or boundary path usage are retained in output but excluded from Gaussian testing.
5. Fit mouse-replicate GLS models with separate inferential and between-mouse variance components. Test condition within cell type, and test cell type within mouse, as separate analyses.

Two inferential covariance calculations are reported. The *conditional* calculation fixes the fitted transcript mixture within each path. The *profile* calculation allows all transcript directions to vary as nuisance parameters. They are sensitivity analyses for the same block-by-cell-type hypothesis, not independent biological hypotheses.

Statistical Model

Paired-primer EC likelihood

For pseudobulk i and primer p, let $\boldsymbol{c}_{ip}$ be the EC counts, $N_{ip} = \mathbf{1}'\boldsymbol{c}_{ip}$, and A_p the fixed weighted EC design. For transcript abundance $\boldsymbol{\theta}_i$,

$$\boldsymbol{q}_{ip} = A_p\boldsymbol{\theta}_i, \qquad Z_{ip} = \mathbf{1}'\boldsymbol{q}_{ip},$$

and the likelihood conditional on the primer-specific UMI total is

$$\boldsymbol{c}_{ip} \mid N_{ip}, \boldsymbol{\theta}_i \sim \text{Multinomial}\left(N_{ip}, \frac{\boldsymbol{q}_{ip}}{Z_{ip}}\right).$$

Oligo(dT) and random-hexamer counts have separate designs and likelihood terms, preserving their different positional and transcript-exposure profiles.

The genome-wide hierarchical model supplies a pseudobulk baseline $\boldsymbol{\theta}_i^{(0)}$. For block b, a local parameter $\boldsymbol{\delta}_{bi}$ changes the mass of all transcripts carrying path $s(t)$:

$$\theta_{it}(\boldsymbol{\delta}_{bi}) \propto \theta_{it}^{(0)} \exp\{Q'_{b,s(t)}\boldsymbol{\delta}_{bi}\}.$$

Here Q_b is a Helmert basis for the $S_b - 1$ dimensional sum-zero subspace. The total spliced mass and transcript mixture within each path are fixed during this local refinement.

Cell-resolved likelihood

The cell-resolved alternative tests whether aggregation discards useful read-assignment information. Let c index cells in pseudobulk i, and let $\boldsymbol{\theta}_{ci}^{(0)}$ be the cell's hierarchical transcript baseline. One path perturbation is shared by all cells in the pseudobulk:

$$\theta_{cit}(\boldsymbol{\delta}_{bi}) \propto \theta_{cit}^{(0)} \exp\{Q'_{b,s(t)}\boldsymbol{\delta}_{bi}\}.$$

The proportionality is normalized separately within each cell's block transcripts, preserving its baseline block mass. The objective is the sum of cell- and primer-specific multinomial log likelihoods,

$$\ell_{bi}(\boldsymbol{\delta}) = \sum_{c \in i} \sum_{p} \log \text{Multinomial}\left(\boldsymbol{c}_{cip}; N_{cip}, \frac{A_p\boldsymbol{\theta}_{ci}(\boldsymbol{\delta})}{\mathbf{1}'A_p\boldsymbol{\theta}_{ci}(\boldsymbol{\delta})}\right).$$

Thus cells contribute read-assignment information without introducing cell-level biological degrees of freedom.

The reported pseudobulk path proportions average fitted cell proportions using predicted gene-molecule weights,

$$w_{ci} = (\text{cell total UMI})_{ci} \times (\text{decoded gene proportion})_{ci}.$$

The expected information for $\boldsymbol{\delta}_{bi}$ is summed across cells and primers. Its inverse is propagated through this weighted path average to obtain the covariance of $Q_b' \log \boldsymbol{\psi}_{bi}$. The downstream mouse-replicate REML–GLS model is then unchanged.

Path response and inferential covariance

Let C_b map transcripts to paths. Path proportions and the response used by the differential test are

$$\boldsymbol{\psi}_{bi} = C_b \frac{\boldsymbol{\theta}_{i,\text{spliced}}}{\mathbf{1}'\boldsymbol{\theta}_{i,\text{spliced}}}, \qquad \boldsymbol{y}_{bi} = Q_b' \log \boldsymbol{\psi}_{bi}.$$

For two paths, y_{bi} is the log path-odds divided by $\sqrt{2}$. Under the local parameterization, $\boldsymbol{y}_{bi} = \boldsymbol{y}_{bi}^{(0)} + \boldsymbol{\delta}_{bi}$.

The conditional covariance is the inverse Fisher information for $\boldsymbol{\delta}_{bi}$. For the profile calculation, write $\boldsymbol{\eta}_i = \log \boldsymbol{\theta}_i$. The expected transcript information is

$$I_{\eta,ip} = N_{ip} \left[\frac{\operatorname{diag}(\boldsymbol{\theta}_i) A_p' \operatorname{diag}(\boldsymbol{q}_{ip}^{-1}) A_p \operatorname{diag}(\boldsymbol{\theta}_i)}{Z_{ip}} - \boldsymbol{r}_{0,ip} \boldsymbol{r}_{0,ip}' \right],$$

where $\boldsymbol{d}_p = A_p' \mathbf{1}$ and $\boldsymbol{r}_{0,ip} = (\boldsymbol{d}_p \odot \boldsymbol{\theta}_i)/Z_{ip}$. Information is added across primers. If $J_{bi} = \partial \boldsymbol{y}_{bi} / \partial \boldsymbol{\eta}_i'$, then

$$V_{bi}^{\text{profile}} = J_{bi} \left(\sum_p I_{\eta,ip} \right)^{+} J_{bi}'.$$

The implementation checks whether J_{bi} overlaps the information null space. A non-estimable contrast is reported as unidentified rather than being assigned zero variance by a pseudoinverse.

Dirichlet–multinomial alternative

A compositional likelihood provides an alternative to Gaussian ILR regression. It must account for EC ambiguity: treating all gene UMIs as known path assignments is overconfident. For fitted path proportions $\boldsymbol{\psi}_i$, define the unit-multinomial ILR covariance

$$B_i = Q' \operatorname{diag}(\boldsymbol{\psi}_i^{-1}) Q.$$

The effective multinomial size matches B_i/n_i to the EC-derived conditional covariance V_i in Frobenius norm:

$$n_i^{\text{eff}} = \frac{\langle B_i, B_i \rangle}{\langle B_i, V_i \rangle}, \qquad n_i^{\text{eff}} \le \text{gene UMIs}_i.$$

Fractional path counts are $\boldsymbol{x}_i = n_i^{\text{eff}} \boldsymbol{\psi}_i$.

The compositional regression is

$$\boldsymbol{x}_i \mid n_i^{\text{eff}} \sim \text{DirichletMultinomial}(n_i^{\text{eff}}, \kappa \boldsymbol{\mu}_i), \qquad \operatorname{logit}_{\text{ref}}(\boldsymbol{\mu}_i) = X_i B.$$

The primary analysis estimates the concentration κ under the null and holds it fixed under the alternative, parallel to estimating biological variance under the GLS null. A sensitivity mode re-estimates one global concentration under the alternative. This adds no likelihood-ratio degree of freedom because both models contain one concentration nuisance parameter. Both modes explicitly evaluate the exact multinomial endpoint $\kappa = \infty$, rather than relying on a large finite approximation. Analytic gradients are used for both likelihoods.

Mouse-replicate differential tests

Within one cell type and block, let i index mouse pseudobulks and let X_i encode condition:

$$\widehat{\boldsymbol{y}}_{bi} = X_i B_b + \boldsymbol{\epsilon}_{bi}, \qquad \mathrm{Var}(\boldsymbol{\epsilon}_{bi}) = V_{bi} + \tau_b^2 I.$$

The EC-derived V_{bi} measures inferential uncertainty. The scalar τ_b^2 measures residual between-mouse variation and is estimated by REML under the relevant null model. GLS uses $(V_{bi} + \tau_b^2 I)^{-1}$.

For the *condition test*, one model is fit for each block and cell type. X_i contains an intercept and condition indicators, and the Wald statistic tests all non-reference condition coefficients. Condition labels are permuted with their group sizes fixed for calibration.

For the primary *cell-type test*, the effective path counts are fit by Dirichlet–multinomial regression. The null design contains a fixed effect for each mouse and the alternative adds cell-type indicators:

$$H_0 : B_{\mathrm{cell}} = 0, \qquad H_1 : B_{\mathrm{cell}} \neq 0.$$

Because condition is constant within a mouse, the mouse effects also absorb condition and all other shared mouse-level shifts. The likelihood-ratio statistic jointly tests every non-reference cell type and every path log-ratio. Cell-type labels are shuffled within mouse, preserving pairing, mouse-specific label availability, and pseudobulk sizes. The primary p-value is the finite-permutation rank

$$p_b = \frac{1 + \sum_{r=1}^{R} \mathbb{1}\{T_b^{(r)} \geq T_b\}}{R+1}, \qquad R = 500.$$

The null fit and its concentration are reused, but each permuted alternative is refit. Statistics within relative tolerance 10^{-6} are counted as ties in the permutation rank; this prevents optimizer noise at likelihood boundaries from changing exact p-values. Benjamini–Hochberg correction is then applied across the nonredundant partitions passing the observation-count filter. The earlier conditional- and profile-covariance Wald tests are retained as sensitivity analyses.

Joint condition tests

Three joint tests were evaluated. First, ACAT combines cell-type-specific p-values for one path partition:

$$p_{\mathrm{ACAT}} = \frac{1}{2} - \frac{1}{\pi} \arctan\left[\frac{1}{J}\sum_{j=1}^{J} \tan\{\pi(1/2 - p_j)\}\right].$$

This remains valid under dependent component p-values and does not assume a common effect direction. A second ACAT level combines distinct path partitions within a gene.

Second, a shared-effect model uses cell-type-specific intercepts and one condition effect across cell types. Treating the resulting cell-type pseudobulks as independent is an explicitly evaluated negative control. The valid Gaussian version includes mouse clustering:

$$\boldsymbol{y}_{mc} = X_{mc} B + \boldsymbol{u}_m + \boldsymbol{e}_{mc},$$

$$\mathrm{Cov}(\boldsymbol{y}_{mc}, \boldsymbol{y}_{mc'}) = \tau_m^2 I \quad (c \neq c'), \qquad \mathrm{Var}(\boldsymbol{y}_{mc}) = V_{mc} + (\tau_m^2 + \tau_r^2) I.$$

The shared-mouse variance τ_m^2 and cell-type residual variance τ_r^2 are estimated by two-component REML under the no-condition model. Condition labels are permuted once per biological mouse and applied to all of its cell types.

Implementation Details

Splice-block construction

GTF exons are represented by zero-based, half-open intervals. Exon boundaries define atomic intervals; intervals exonic in every annotated spliced transcript are merged into constitutive anchors. Consecutive anchors define a block, and a transcript path is its ordered exon-interval chain between them. Terminal pseudo-anchors include alternative first and last regions. A gene without a shared constitutive region receives one whole-gene block.

Blocks with one path are discarded. Simple two-path blocks include skipped exons, mutually exclusive exons, alternative donors and acceptors, and retained introns. The implementation also supports nested and multi-path blocks. Precursor features remain nuisance features in the EC model but are not assigned to spliced paths.

Different genomic blocks can induce exactly the same partition of annotated transcripts, for example when several exons always occur together. Such blocks have identical statistical responses and must not count as repeated evidence. A canonical partition is the unordered collection of transcript sets assigned to paths. Equivalent blocks are collapsed before FDR or joint testing; all member interval identifiers remain in the output.

Pseudobulks and local fitting

Cell transcript proportions from the hierarchical decoder are weighted by cell UMI totals and aggregated by mouse, condition, and cell type. Pseudobulks must contain at least 20 cells and 10^5 total UMIs. A block is fit when its gene has at least 25 gene-assigned UMIs in at least two qualifying pseudobulks. Only ECs whose compatible transcripts map uniquely to that gene enter the local likelihood.

Bounded L-BFGS optimizes the paired multinomial likelihood, with $|\delta| \leq 20$ to prevent floating-point underflow. Two-path fits are initialized from a log-ratio grid because the paired-primer normalizers can make the objective non-concave.

For the cell-resolved variant, sparse cell-by-EC matrices are retained and one local transcript matrix is decoded at a time. The optimizer dimension is still only the number of paths minus one. Cell-specific Fisher contributions are accumulated analytically, so the method does not fit one free splicing parameter per cell.

Identifiability and reliability

Finite Fisher covariance is not sufficient near a simplex boundary. Parametric bootstrap identified two reliability requirements:

$$\lambda_{\max}(V_{bi}) \leq 1/8 \quad \text{and} \quad \min_s \widehat{\psi}_{bis} \geq 0.15.$$

Thus every tested log-ratio direction has effective information of at least eight and every fitted path has at least 15% usage. All estimates are written, together with convergence, identifiability, and reliability flags; only reliable estimates enter GLS.

Results

Condition-test scope

The primary analysis used the 25-gene-UMI cutoff and blocks with at most eight represented paths. It fit 10,100 blocks and 359,713 pseudobulk-by-block path estimates; 359,711 fits converged. After reliability filtering, there were 829 conditional and 506 profile tests.

These 1,335 rows represent 829 distinct block-by-cell-type hypotheses, covering 527 distinct splice blocks in eight cell types. The profile analysis is available for a subset of the conditional hypotheses. The possible conditions were 5xFAD, 5xFIRE, 5xFIRE PBS, 5xFIRE transplant, FIRE, and WT. For the 829 conditional hypotheses, 275 compared two conditions, 155 compared three, 153 compared four, 130 compared five, and 116 compared all six. The included set varies because every condition in a test must have at least three independent mouse pseudobulks for that block and cell type.

Condition-test results

No block was significant at 5% FDR in either covariance analysis. At the unadjusted $p < 0.05$ level:

- the conditional analysis identified 24 block-by-cell-type hypotheses, covering 21 distinct blocks;
- the profile analysis identified 20 block-by-cell-type hypotheses, covering 17 distinct blocks;
- 20 hypotheses were nominally significant in both analyses.

The union is 24 nominal block-by-cell-type hypotheses covering 21 blocks, but the corrected number of significant hypotheses and blocks is zero. Thus there is no FDR-significant evidence for condition-dependent splicing within cell type. The two covariance variants should not be added as if they were separate discoveries.

Cell-type effects within mouse

The initial Gaussian analysis tested 369 conditional and 218 profile partitions. It found 22 and five FDR-significant partitions, respectively, with all profile discoveries contained in the conditional set. After canonicalizing annotation intervals that encode the same transcript-to-path partition, these 22 intervals represented 14 distinct partitions.

The compositional analysis used converged, conditionally identifiable estimates and required every modeled path proportion to be at least 5%. This produced 646 nonredundant tests. A minimum of 30 pseudobulk observations per partition, chosen as a label-effect-independent coverage requirement, retained 406 hypotheses for FDR correction. Among these, 149 had exact permutation $p < 0.05$, and **101 splice-path partitions in 95 genes passed 5% FDR**. Thirteen of the 14 nonredundant Gaussian discoveries were recovered; the remaining partition had exact $p = 0.032$ but FDR 0.098. Significant compositional results had median maximum cell-type logit effect 0.84 and median effective path count 21.7.

The observation-count threshold is not a narrow boundary effect: thresholds from 29 through 32 pseudobulks each gave 101 discoveries, while 33 gave 99. All 646 observed fits converged. The median number of valid permutations was 500; the minimum was 476 after excluding failed permutation refits. Asymptotic Dirichlet–multinomial p-values were anti-conservative, with 15.5% of permutation-null values below 0.05, which is why only exact per-event permutation ranks are used for the primary result.

Two sensitivity analyses did not improve the event-level result. Testing 3,708 individual cell-type pairs gave 97 FDR-significant pair hypotheses but only 32 distinct partitions after correction across all pairs. Restricting each block to its two molecule-weighted dominant paths increased coverage; with a 1% path floor it gave 88 discoveries, and removing the floor gave 91, under a cross-fitted empirical null. These analyses support the full-path exact omnibus test as the primary cell-type analysis.

Allowing the alternative to re-estimate κ also left the result unchanged: it found the same 101 partitions in the same 95 genes, with no event entering or leaving the 5% FDR set. Exact p-values from fixed- and free-concentration fits had Spearman correlation 0.9995, and both had 148 nominal events. The null optimum was multinomial for 499 of 646 tests, and the alternative optimum was multinomial for 580. Among 66 tests with finite concentration in both models, the median alternative/null concentration ratio was 1.54, consistent with cell-type mean effects explaining some apparent overdispersion. Exact permutation calibration absorbs this nuisance-parameter flexibility, so freeing κ provides no net power gain here.

Calibration

Across the 25-UMI analysis, 2.7% of permuted-label Wald p-values were below 0.05 for both conditional and profile covariance, indicating conservative null calibration. A parametric bootstrap of 3,000 reliable local fits gave mean squared standardized error 1.10 and 94.5% coverage for nominal 95% intervals. Independent Gaussian simulations of REML–GLS gave type-I error between 2.7% and 5.0% across the tested biological-variance settings.

Numerical tests additionally verify skipped-exon block construction, identifiable covariance after collapsing indistinguishable transcripts, rejection of information-null path contrasts, the analytic binomial Fisher variance, and recovery of imposed GLS effects.

Coverage-cutoff sensitivity

UMIs	blocks	tests	conditional null $p < .05$	profile null $p < .05$	bootstrap coverage
25	10100	1335	0.027	0.027	0.945
50	5896	755	0.031	0.030	0.936
100	2820	283	0.029	0.025	0.946
200	1034	69	0.028	0.026	0.948
500	195	10	0.030	0.050	0.946

No cutoff produced an FDR-significant block. The 500-UMI calibration estimate is based on only 100 permutation tests.

For reliable two-condition, two-path conditional tests, mean asymptotic power for a 0.5 ILR effect was 0.49, 0.51, 0.44, 0.56, and 0.59 at increasing cutoffs. The final two values are selected from only eight and two eligible tests. Among 103 hypotheses shared by the 25- and 50-UMI analyses, power was 0.453 and 0.449. Lower coverage cutoffs therefore increase the number of testable blocks without degrading calibrated precision among shared tests. The 25-UMI cutoff is the default.

Does cell resolution increase condition-test power?

The cell-resolved likelihood was run on the 527 blocks testable in the 25-UMI primary analysis. All 41,242 block-by-pseudobulk fits converged; 21,624 passed the reliability rules, compared with

21,010 pseudobulk conditional estimates on the same blocks. This produced 836 condition tests, compared with 829 for pseudobulk fitting. No cell-resolved test passed 5% FDR. Condition-label permutations were conservative, with 2.49% of null p-values below 0.05.

An exact matched comparison used the same reliable samples for both methods in 807 block-by-cell-type tests. Cell resolution reduced the median local covariance trace by 1.6%, and the median path-response RMS difference was 0.033 ILR units. P-values were strongly concordant (Spearman $\rho = 0.958$); 24 pseudobulk and 25 cell-resolved tests were nominally significant.

For the 259 matched two-condition, two-path tests, mean asymptotic power for a 0.5 ILR effect was 0.504 for both methods. The median ratio of cell-resolved to pseudobulk condition-coefficient standard error was 0.995. Retaining cell likelihood terms therefore did not materially increase power in this dataset. The aggregated EC counts and decoder baseline preserve nearly all relevant local information within a cell type, while residual between-mouse variation dominates the small reduction in inferential variance. Pseudobulk fitting remains the default; the cell-resolved method is available for datasets with stronger within-pseudobulk transcript heterogeneity.

Compositional and joint-test assessment

Collapsing equivalent transcript partitions reduced the 829 conditional block-by-cell-type rows to 505 nonredundant tests over 317 partitions. The median effective path count was 32 molecules, or 62% of gene UMIs; only 0.6% of estimates reached the gene-UMI cap.

On the same 505 tests, conditional GLS had 12 nominal results and the Dirichlet–multinomial had 18. Neither had a 5% FDR discovery. The Dirichlet–multinomial was conservatively calibrated: 3.76% of 10,100 permuted-label p-values were below 0.05. In contrast, pure multinomial regression gave 197 nominal and 155 FDR results, but 39.1% of its permutation p-values were below 0.05. UMI sampling alone therefore cannot represent the large between-mouse compositional variation. Output tables mark pure multinomial and naive shared-effect rows as invalid for inference while retaining them as calibration diagnostics.

ACAT across cell types produced 8 nominal GLS and 12 nominal Dirichlet–multinomial partitions among 317 tests, with no FDR discoveries. Combining distinct partitions within genes gave 7 and 11 nominal genes among 276 tests, again with no FDR discoveries.

Naive shared-effect pooling illustrates the pseudoreplication risk. It gave 43 nominal and one FDR Dirichlet–multinomial partition, but synchronized mouse permutations rejected 13.2% of nulls at 0.05. The apparent discovery had asymptotic $p = 6.4 \times 10^{-5}$ but permutation $p = 0.143$. Naive shared GLS was also inflated, with 8.4% null rejection.

The mouse-clustered shared GLS restored conservative calibration: 2.19% of 7,169 synchronized permutation p-values were below 0.05. It produced 11 nominal partitions among 359 tests and no FDR discoveries; its smallest p-value was 0.0126 with FDR 0.86. Gene-level ACAT gave 9 nominal genes among 313 and no FDR discoveries. Thus the calibrated gain is limited to the increase from 12 to 18 nominal within-cell-type tests under the Dirichlet–multinomial. None of the evaluated strategies creates a genome-wide condition discovery in this dataset.

Limitations

The profile covariance is conditional on the fitted genome-wide decoder. Uncertainty in shared decoder parameters is expected to be small for common paths but may matter for rare genes or cell types. A mouse-level jackknife of the hierarchical fit is the preferred extension.

The current biological variance is isotropic in log-ratio space. Estimating a full covariance separately for every block would be unstable with few mice. A later empirical-Bayes model could shrink path-specific biological covariances across genes.